

India's Agricultural Crop Production Analysis.

TEAM ID	SWTID-2026-8827
PROJECT TITLE	India's Agricultural Crop Production Analysis.
MAX MARKS	5 MARKS

4.3 – Solution Architecture

1. Data Sources

- Collect crop production data from government portals, CSV files, Excel files, and open datasets.
- Gather information such as State, District, Crop, Season, Year, Area, Production, and Yield.

2. Data Collection

- Import the agricultural datasets into the Tableau environment.
- Combine data from multiple sources into a single dataset.

3. Data Preparation

- Remove duplicate records and missing values.
- Correct data types and rename columns.
- Standardize and clean the data for accurate analysis.

4. Data Processing

- Create calculated fields and measures.
- Build relationships between different data tables.
- Aggregate production and yield values for analysis.

5. Data Visualization

- Develop interactive dashboards using Tableau.
- Create bar charts, line charts, maps, treemaps, pie charts, and KPI cards.

6. Interactive Analysis

- Add filters for State, Crop, Season, and Year.
- Enable drill-down functionality for detailed exploration of data.

7. Insights Generation

- Analyze crop production trends over time.
- Compare state-wise agricultural performance.
- Identify high-production and low-production regions.

8. Reporting and Decision Support

- Generate dynamic reports and dashboards.
- Support farmers, researchers, and policymakers with data-driven insights.
- Improve agricultural planning and resource allocation.

9. Final Outcome

- A centralized Tableau dashboard that provides accurate, interactive, and easy-to-understand agricultural crop production analysis for better decision-making.